EXECUTIVE RECOMMENDATION | CASE STUDY

NEXORA FRAUD DETECTION

Prioritize accounts younger than 7 days for manual review

75%
Fraud recall in validation

18.75%
Alert precision

16/day
Alerts vs 40/day capacity



Vitoria Rodrigues Miranda

Prioritize new accounts to capture 75% of confirmed fraud

RECOMMENDATION

Pilot a manual-review rule for transactions from accounts younger than seven days.

75%

Fraud recall in validation

18.75%

Alert precision

16/day

Daily alerts vs 40/day capacity

Evidence: simulated validation, days 9–12. Not a forecast of future performance.

Nexora can review only half of the 80 transactions it processes each day

SITUATION

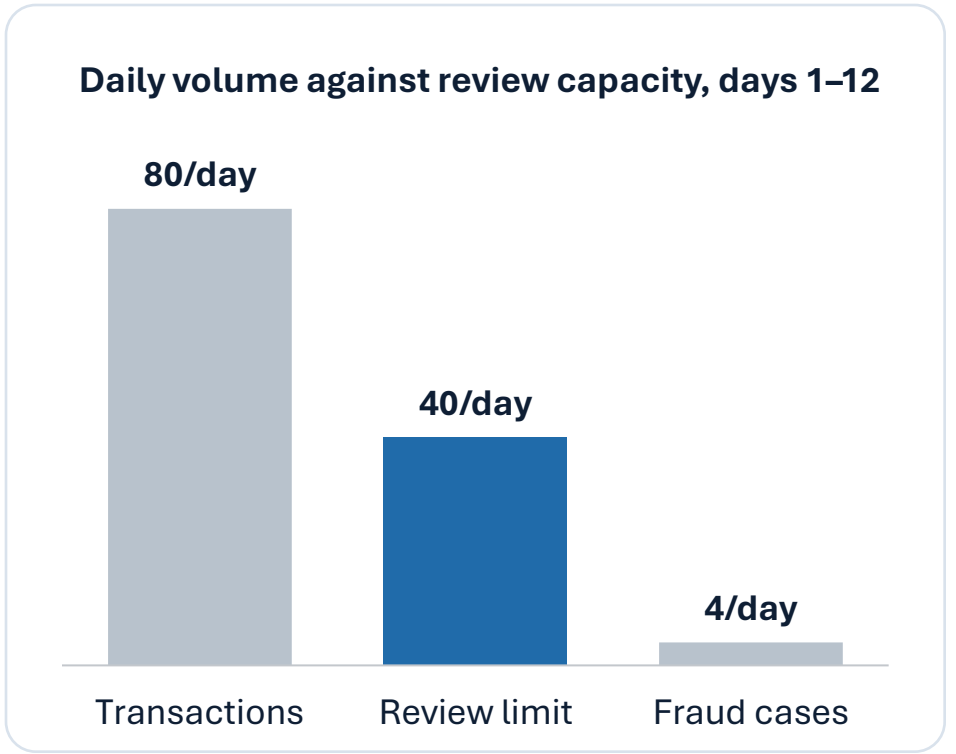
Nexora processes 80 transactions a day; 5% are confirmed fraud.

COMPLICATION

The review team can handle at most 40 cases a day, so it cannot check every transaction.

QUESTION

Which transactions should Nexora review first to catch more fraud within its daily limit?



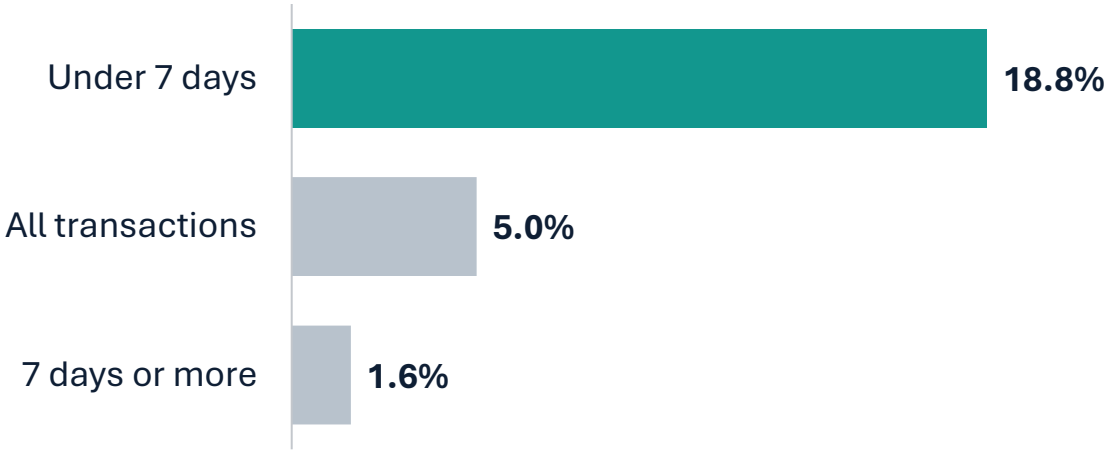
DECISION CRITERIA

- Fraud recall
- Precision
- Alerts per day
- Operational simplicity

Accounts under 7 days old show an 18.8% fraud rate, 12× that of older accounts

EXPLORATION: DAYS 1–8

Fraud rate by account age, days 1–8



New accounts are **20% of transactions** but **75% of confirmed fraud.**

Account age < 7 days

18.8% fraud rate vs 1.6% for older accounts; selected for validation

Prior device sharing

Older accounts: 1.4% fraud on shared devices vs 2.1% on unshared

30-minute device velocity

No qualifying events among older accounts in exploration

Exploration identified candidates; the rule choice is supported by subsequent temporal validation.

The rule flagged 12 of 16 frauds while using 40% of daily review capacity

VALIDATION: DAYS 9–12

RECALL

75%

12 of 16 confirmed frauds flagged; **4 missed**

PRECISION

18.75%

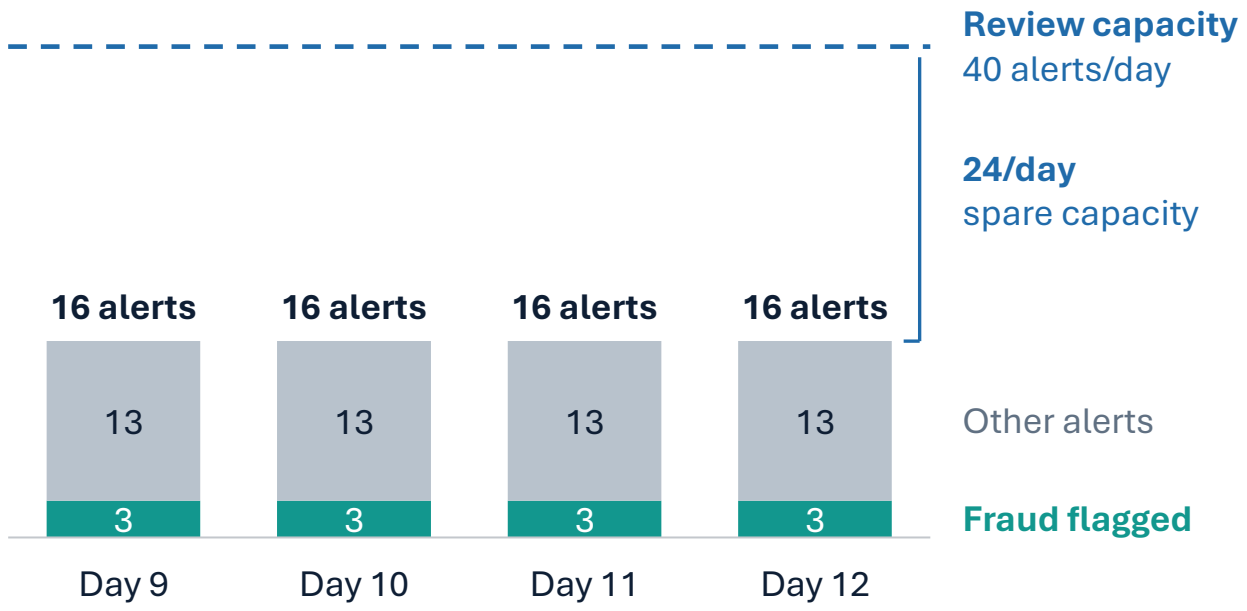
12 frauds in 64 alerts, 3.75× the 5% base rate

ALERT LOAD

16/day

40% of the 40-alert daily review capacity

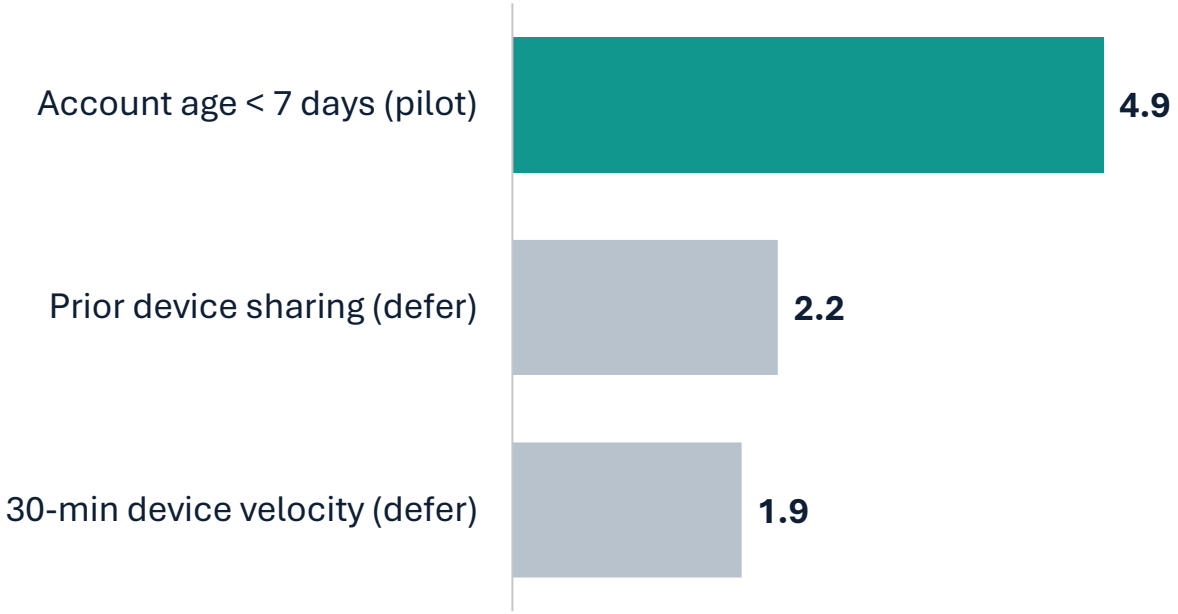
Daily alerts by outcome against review capacity, days 9–12



Simulated data, 4-day validation window (days 9–12). Results are descriptive, not a forecast.

Device-based signals do not justify inclusion in the initial rule

Weighted decision score by signal, out of 5



WHY DEVICE SIGNALS ARE DEFERRED

1.4% vs 2.1%

Fraud rate on shared vs unshared devices among older accounts, days 1–8

0

Extra alerts from 30-minute device velocity among older accounts, days 9–12

Illustrative weighted scores: impact 40%, operational feasibility 30%, ease 20%, evidence 10%.

Launch a controlled pilot and investigate missed fraud

01	**Pilot the account-age rule**	Manual review; no automatic blocking
02	**Monitor performance**	Track alerts/day, precision and recall
03	**Investigate missed cases**	Analyze the four unflagged fraud cases
04	**Refine and revalidate**	Test incremental signals on new data

Decision requested: approve a monitored manual-review pilot, subject to operational checks.

Appendix | Evidence, assumptions and limitations

Observed validation evidence

- 4-day temporal holdout (days 9–12)
- 16 confirmed frauds; 12 flagged; 4 missed
- 64 alerts total; 16/day average
- 75% recall; 18.75% precision
- 30-minute velocity: no incremental alerts among older accounts

Fraud by account age and window

Window	Account age	Txns	Fraud	Fraud rate
Days 1–8	Under 7 days	128	24	18.8%
Days 1–8	7 days or more	512	8	1.6%
Days 9–12	Under 7 days	64	12	18.8%
Days 9–12	7 days or more	256	4	1.6%
Days 1–12	All accounts	960	48	5.0%

Limitations

Small simulated sample; no measured implementation cost, prevented losses, or future performance. Weighted scores are managerial estimates, not statistical outputs.

Appendix | Source data

SOURCE DATA

Nexora_Fraud_Validation.xlsx

Full transaction data and the calculations behind every figure in this deck.

Open the source spreadsheet : Google Sheets

EXPLORATION & VALIDATION

Fraud rate by account age, days 1–8 and 9–12, plus rule metrics

DEVICE SIGNALS

Prior device sharing and 30-minute device velocity tests

TRANSACTIONS

960 simulated transactions over 12 days, one row each